

CohortConstructor

Build and Curate Study Cohorts in R Using the OMOP Common Data Model

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Introduction

- CohortConstructor package is designed to support cohort building pipelines in R, using data mapped to the OMOP Common Data Model.
- The code is publicly available in OHDSI's GitHub repository **CohortConstructor**.
- CohortConstructor v0.6.3 is available in **CRAN**.
- Vignettes with further information can be found in the package **website**.

Set up

OMOP Cohorts in R

OMOP Cohorts in R

- The `<cohort_table>` class is defined in the R package **omopgenerics**.
- This is the class that **CohortConstructor** uses, as well as other OMOP analytical packages.
- As defined in **omopgenerics**, a `<cohort_table>` must have at least the following 4 columns (without any missing values in them):
 - **cohort_definition_id**: Unique identifier for each cohort in the table.
 - **subject_id**: Unique patient identifier.
 - **cohort_start_date**: Date when the person enters the cohort.
 - **cohort_end_date**: Date when the person exits the cohort.

OMOP Cohorts in R

```
cdm$cohort
```

```
# A query:  ?? x 4
# Database: DuckDB 1.5.5 [unknown@Linux 6.17.0-1022-azure:R 4.6.1//tmp/Rtmp9NTze3/file2097325bf3c7.duckdb]
  cohort_definition_id subject_id cohort_start_date cohort_end_date
                <int>      <int> <date>              <date>
1                   1        123 1988-11-28            1988-12-07
2                   1        263 2002-12-16            2002-12-24
3                   1       1384 1987-09-20            1987-09-28
4                   1       1627 1965-04-26            1965-05-08
5                   1       1796 2005-07-10            2005-07-20
6                   1      4532 1939-02-22            1939-03-03
7                   1      4855 2008-03-11            2008-03-22
8                   2        202 2010-03-04            2010-03-25
9                   2        437 1965-03-03            1965-03-24
10                  2       640 1998-05-02            1998-05-16
... 
```

OMOP Cohorts in R

Additionally, the `<cohort_table>` object has the following **attributes**:

- **Settings**: Relate each cohort definition ID with a cohort name and other variables that define the cohort.

```
settings(cdm$cohort)
```

```
# A tibble: 2 × 4
  cohort_definition_id cohort_name      cdm_version vocabulary_version
      <int> <chr>                <chr>          <chr>
1           1 viral_pharyngitis 5.3          v5.0 18-JAN-19
2           2 viral_sinusitis  5.3          v5.0 18-JAN-19
```

OMOP Cohorts in R

- **Attrition:** Store information on each inclusion criteria applied and how many records and subjects were kept after.

```
attrition(cdm$cohort)
```

```
# A tibble: 8 × 7
  cohort_definition_id number_records number_subjects reason_id reason
excluded_subjects
      <int>          <int>          <int>    <int> <chr>
<int>
1          1        10217         2606      1 Initial qualifying events
0
2          1        10217         2606      2 Record in observation
0
3          1        10217         2606      3 Not missing record date
0
4          1        10217         2606      4 Merge overlapping records
0
5          2        17268         2686      1 Initial qualifying events
-
```

OMOP Cohorts in R

- **Cohort count:** Number of records and subjects for each cohort.

```
cohortCount(cdm$cohort)
```

```
# A tibble: 2 × 3
```

	cohort_definition_id	number_records	number_subjects
	<int>	<int>	<int>
1	1	10217	2606
2	2	17268	2686

OMOP Cohorts in R

- **Cohort codelist:** Codelists used to define entry events and inclusion criteria for each cohort.

```
attr(cdm$cohort, "cohort_codelist")
```

```
# A query: ?? x 4
# Database: DuckDB 1.5.5 [unknown@Linux 6.17.0-1022-azure:R 4.6.1//tmp/Rtmp9NTze3/file2097325bf3c7.duckdb]
  cohort_definition_id codelist_name      concept_id codelist_type
      <int> <chr>              <int> <chr>
1             1 viral_pharyngitis    4112343 index event
2             2 viral_sinusitis     40481087 index event
```

CohortConstructor function Sets

Base cohorts *Cohort*

construction based on clinical concepts or demographics.

Requirements and Filtering

Demographic restrictions, event presence/absence conditions, and filtering specific records.

Update cohort entry and exit

Adjusting entry and exit dates to align with study periods, observation windows, or key events.

Transformation and

Combination *Merging, stratifying, collapsing, matching, or intersecting cohorts. (Not covered today)*

Base cohorts

Functions to build base cohorts

- `demographicsCohort()`
- `conceptCohort()`
- `measurementCohort()`
- `deathCohort()`

Create the cdm_reference object

```
cdm <- omock::mockCdmFromDataset(datasetName = "GiBleed",  
                                  source = "duckdb")
```

Demographics based - Example

- Two cohorts, females and males, both aged 18 to 60 years old, with at least 365 days of previous observation in the database.

```
cdm$age_cohort <- demographicsCohort(  
  cdm = cdm,  
  ageRange = c(18, 60),  
  sex = c("Female", "Male"),  
  minPriorObservation = 365,  
  name = "age_cohort"  
)  
  
settings(x = cdm$age_cohort)
```

A tibble: 2 × 5

	cohort_definition_id	cohort_name	age_range	sex	min_prior_observation
	<int>	<chr>	<chr>	<chr>	<dbl>
1	1	demographics_1	18_60	Female	365
2	2	demographics_2	18_60	Male	365

Demographics based - Example

```
cohortCount(cohort = cdm$age_cohort)
```

```
# A tibble: 2 × 3
  cohort_definition_id number_records number_subjects
      <int>           <int>           <int>
1             1         1373         1373
2             2         1321         1321
```

```
attrition(x = cdm$age_cohort)
```

```
# A tibble: 12 × 7
  cohort_definition_id number_records number_subjects reason_id reason
excluded_records excluded_subjects
      <int>           <int>           <int>      <int> <chr>
<int>      <int>
1             1         2694         2694      1 Initial qualifying events
0             0
2             1         2694         2694      2 Non-missing sex
0             0
3             1         1373         1373      3 Sex requirement: Female
1321        1321
4             1         1373         1373      4 Non-missing year of birth
0             0
5             1         1373         1373      5 Age requirement: 18 to 60
```

Demographics based - Example

To better visualise the **attrition**, we can use the package **CohortCharacteristics** to create a **formatted table**:

```
cdm$age_cohort |>
  summariseCohortAttrition() |>
  tableCohortAttrition()
```

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
GiBleed; demographics_1				
Initial qualifying events	2,694	2,694	0	0
Non-missing sex	2,694	2,694	0	0
Sex requirement: Female	1,373	1,373	1,321	1,321
Non-missing year of birth	1,373	1,373	0	0
Age requirement: 18 to 60	1,373	1,373	0	0
Prior observation requirement: 365 days	1,373	1,373	0	0
GiBleed; demographics_2				
Initial qualifying events	2,694	2,694	0	0
Non-missing sex	2,694	2,694	0	0
Sex requirement: Male	1,321	1,321	1,373	1,373
Non-missing year of birth	1,321	1,321	0	0
Age requirement: 18 to 60	1,321	1,321	0	0
Prior observation requirement: 365 days	1,321	1,321	0	0

Concept based - Example

Let's create a cohort of medications that contains two drugs: diclofenac, and acetaminophen.

1. Create codelist of the drug of interest

```
drugCodes <- list(  
  "diclofenac" = c(1124300L),  
  "acetaminophen" = c(1125315L, 1127078L, 1127433L, 19133768L, 40162522L, 40229134L, 40231925L)  
)
```

Concept based - Example

2. Create concept based cohorts

```
cdm$medications <- conceptCohort(  
  cdm = cdm,  
  conceptSet = drugCodes,  
  name = "medications"  
)
```

```
settings(x = cdm$medications)
```

```
# A tibble: 2 × 4
```

	cohort_definition_id	cohort_name	cdm_version	vocabulary_version
	<int>	<chr>	<chr>	<chr>
1	1	acetaminophen	5.3	v5.0 18-JAN-19
2	2	diclofenac	5.3	v5.0 18-JAN-19

Concept based - Example

- Attrition

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
diclofenac				
Initial qualifying events	850	850	0	0
Record in observation	830	830	20	20
Not missing record date	830	830	0	0
Merge overlapping records	830	830	0	0

Concept based - Example

- Cohort codelist as an attribute

```
attr(cdm$medications, "cohort_codelist")
```

```
# A query: ?? x 4
# Database: DuckDB 1.5.5 [unknown@Linux 6.17.0-1022-azure:R 4.6.1//tmp/Rtmp9NTze3/file20975e56d1.duckdb]
  cohort_definition_id codelist_name concept_id codelist_type
      <int> <chr>           <int> <chr>
1             1 acetaminophen   1125315 index event
2             1 acetaminophen   1127078 index event
3             1 acetaminophen   1127433 index event
4             1 acetaminophen  19133768 index event
5             1 acetaminophen   40162522 index event
6             1 acetaminophen   40229134 index event
7             1 acetaminophen   40231925 index event
8             2 diclofenac       1124300 index event
```



Your turn!



Exercise 1 - Base cohorts

Create a base cohort table named **base_conditions_cohort** that has two cohorts: Rheumatoid Arthritis (RA) and Pneumonia. Summarise cohort counts and attrition. Use the codelist below.

```
codes <- list(  
  "ra" = c(80809L),  
  "pneumonia" = c(255848L)  
)
```



Click to see solution

Requirements and Filtering

Functions to apply requirements and filter

- On demographics
 - `requireDemographics()`
 - `requireAge()`
 - `requireSex()`
 - `requirePriorObservation()`
 - `requireFutureObservation()`
- On cohort entries
 - `requireIsFirstEntry()`
 - `requireIsLastEntry()`
 - `requireIsEntry()`
- Require presence or absence based on other cohorts, concepts, and tables
 - `requireCohortIntersect()`
 - `requireConceptIntersect()`
 - `requireTableIntersect()`
- Other
 - `requireInDateRange()`
 - `requireMincohortCount()`
 - `requireDuration()`

Requirement functions - Example

- We can apply different inclusion criteria using CohortConstructor's functions in a pipe-line fashion. For instance, in what follows we require
 - only first record per person
 - subjects 18 years old or more at cohort start date
 - only females
 - at least 30 days of prior observation at cohort start date

```
cdm$medications_requirement <- cdm$medications |>
  requireIsFirstEntry(name = "medications_requirement") |>
  requireDemographics(
    ageRange = list(c(18, 150)),
    sex = "Female",
    minPriorObservation = 30,
    name = "medications_requirement"
  )
```

Requirement functions - Example

Attrition Acetaminophen

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Restricted to first entry	2,679	2,679	11,229	0
Age requirement: 18 to 150	309	309	2,370	2,370
Sex requirement: Female	176	176	133	133
Prior observation requirement: 30 days	176	176	0	0
Future observation requirement: 0 days	176	176	0	0

Requirement functions - Example

- Now, we only want to keep those records for which prescription happened on the same day of a healthcare visit.

```
cdm$medications_requirement <- cdm$medications_requirement |>
  requireTableIntersect(
    tableName = "visit_occurrence",
    indexDate = "cohort_start_date",
    window = c(0, 0),
    intersections = c(1, Inf)
  )
```

Requirement functions - Example

Attrition Acetaminophen

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Restricted to first entry	2,679	2,679	11,229	0
Age requirement: 18 to 150	309	309	2,370	2,370
Sex requirement: Female	176	176	133	133
Prior observation requirement: 30 days	176	176	0	0
Future observation requirement: 0 days	176	176	0	0
In table visit_occurrence between 0 & 0 days relative to cohort_start_date between 1 and Inf	1	1	175	175



Your turn!



Exercise 2 - Requirement and filtering

Create a new cohort named “study_cohort” by applying the following criteria to the base conditions cohort:

- For all cohorts, require that records start after January 1, 1990.
- For the Rheumatoid Arthritis cohort, exclude patients with a history of osteoarthritis prior to or on the day of diagnosis.

```
osteoarthritisCodes <- list("osteoarthritis" = c(80180L))
```

▶  Click to see solution

Update cohort entry and exit

Functions to update cohort start and end dates

- Cohort exit
 - `exitAtObservationEnd()`
 - `exitAtDeath()`
 - `exitAtFirstDate()`
 - `exitAtLastDate()`
- Cohort entry
 - `entryAtFirstDate()`
 - `entryAtLastDate()`
- Trim start and end dates
 - `trimDemographics()`
 - `trimToDateRange()`
 - `trimDuration()`
- Pad start and end dates
 - `padCohortDate()`
 - `padCohortEnd()`
 - `padCohortStart()`

Update cohort entry and exit - Example

- We can trim start and end dates to match demographic requirements.
- For instance, cohort dates can be trimmed so the subject contributes time while:
 - Aged 20 to 40 years old
 - Prior observation of at least 365 days

```
cdm$medications_trimmed <- cdm$medications |>
  trimDemographics(
    ageRange = list(c(20, 40)),
    minPriorObservation = 365,
    name = "medications_trimmed"
  )
```

Update cohort entry and exit - Example

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Non-missing year of birth	13,908	2,679	0	0
Age requirement: 20 to 40	4,049	2,124	9,859	555
Prior observation requirement: 365 days	4,049	2,124	0	0



Your turn!



Exercise 3 - Update cohort entry and exit

1) Since Rheumatoid Arthritis is a chronic condition, define its cohort exit so patients remain in the cohort from first diagnostic date to the end of the patient's observation.

👉 Use the ***cohortId*** argument to choose which cohort to modify.

▶ 💡 Click to see solution

CohortConstructor

- 👉 Packages website
- 👉 CRAN link
- 👉 Manual

